

Because, because...

Where does an explanation stop?

A Feynman-inspired investigation of evidence, reduction and the word *why*

Name _____

Date _____

The challenge

You will watch Richard Feynman respond to the question “Why do magnets repel?” Then you will examine one physical explanation he mentions, test it against measurements, and decide how much authority it deserves. The aim is neither to catch out a famous scientist nor to collect the longest possible answer. Your task is to decide:

What does this explanation establish, where does it stop, and how do we know?

A working definition

Reductionism explains a phenomenon by connecting it to smaller parts, lower-level processes, or more general laws. Reduction can deepen an explanation when the links are shown and new predictions follow. It can also change the question or hide relevant context.

Starter: four answers, four different jobs

Two north poles are pushed towards one another. Why do they repel?

- A. They repel because like poles repel.
- B. Each magnet is in the magnetic field produced by the other, so each experiences a force.
- C. Electromagnetic interactions can be represented using a deeper field theory.
- D. They repel so that the magnets do not collide.

Which answer mainly...	A	B	C	D
restates the pattern?	☐	☐	☐	☐
offers a causal model?	☐	☐	☐	☐
uses deeper vocabulary without yet adding a prediction?	☐	☐	☐	☐
smuggles in a purpose?	☐	☐	☐	☐

Your present answer to Feynman’s question:

1 Watch closely: Feynman's chain of "because"

While the clip is playing

Record Feynman's moves rather than trying to transcribe every sentence.

1. What does the interviewer want explained?
2. Why does Feynman resist giving an immediate answer?
3. What does Aunt Minnie help him demonstrate?
4. Where does the story move from an everyday explanation into physics?
5. Listen for the small hedge "*they say*". What work does it do?

Aunt Minnie is in hospital



because she broke her hip



because she fell



because she slipped on ice



because the shoe-ice contact had low friction



because _____

Annotate the ladder. Write one label beside each rung:

everyday convention

physical mechanism

molecular account

still unexplained

An explanation is addressed to someone

For most people, "she is in hospital because she broke her hip" is sufficient. An alien unfamiliar with hospitals would still ask why. Feynman's point is that a why-question stops at facts the questioner and explainer are willing to treat as given.

The reduction question

Would a complete molecular account of ice also explain why Aunt Minnie went to hospital? Explain why the answer is more interesting than a simple yes or no.

Before seeing the research

Feynman's physical suggestion is that pressure under a foot or skate lowers the melting point, creating a lubricating film of water. Record one observation that would support this mechanism and one that might challenge it.

Possible support

Possible challenge

2 The source dossier: authority under examination

Source A: Feynman interview, BBC *Fun to Imagine* (1983)

Claim used in the example: pressure momentarily melts ice beneath a foot, creating a water film.

What is supplied: expert reasoning and an illustration of why-questions. No measurement is presented in the clip. Feynman introduces the physical account with the hedge “they say”.

Source B: Thermodynamic datum

For ordinary ice near 0 °C, increasing pressure by one atmosphere lowers the melting point by approximately
0.0074 °C.

What is supplied: a quantitative relation that allows the proposed mechanism to risk failure.

Source C: Rosenberg, *Physics Today* (2005), research review

Pressure melting is real, although its magnitude is too small to explain ordinary slipperiness across the full temperature range. Ice can remain slippery far below the small melting-point change produced by realistic pressures.

What is supplied: a synthesis comparing the proposed mechanism with scale and observed conditions.

Source D: Surface and gliding research, 2006–2019

Experiments support a disordered, quasi-liquid surface layer on ice below the bulk melting point. Sliding also produces frictional heating. Measurements during gliding indicate that the interfacial film can be much more viscous than ordinary liquid water.

What is supplied: several mechanisms constrained by surface measurements and controlled gliding experiments.

1. Different sources answer different questions

Which source is strongest for...	A	B	C	D
Feynman’s view of why-questions?	☐	☐	☐	☐
calculating a melting-point change?	☐	☐	☐	☐
the state of the scientific case in 2005?	☐	☐	☐	☐
direct evidence about the ice inter-face?	☐	☐	☐	☐

Why is “Which source is most authoritative?” too crude a question?

2. Cross-examine the claim

Claim	Pressure under a foot explains why ice is slippery.
Warrant	Which source supplies evidence rather than reputation? _____
Boundary	Under which temperatures, pressures and types of motion might it work? _____
Adverse evidence	Which observation most weakens the word <i>explains</i> ? _____
Opening	What measurement should researchers make next? _____

Do not flatten the verdict

A mechanism may be physically real and still be too small, too slow, or too restricted to explain the whole phenomenon. **Insufficient** is a more informative judgement than simply **wrong**.

3 Make the explanation risk a number

The model to test

Use the approximate relation

$$\Delta T_{\text{melt}} = (0.0074^\circ\text{C/atm}) \times \Delta P.$$

Treat the numbers below as model inputs. Show units and give sensible precision.

1. A narrow skate contact

Suppose the pressure at part of a skate-ice contact is 150 atmospheres above atmospheric pressure.

- Calculate the predicted lowering of the melting point.
- If the ice is initially at -5.0°C , is pressure melting alone sufficient?

2. Reverse the model

- Estimate the pressure increase required to lower the melting point by 5.0°C .
- Repeat for ice at -20°C .
- State one reason why the calculated pressure is not automatically the pressure beneath every point of a real skate.

3. Grade each proposition separately

Proposition	Secure	Bounded	Not supported
Increasing pressure can lower the melting point of ice.	☐	☐	☐
Pressure melting contributes under some conditions.	☐	☐	☐
Pressure melting explains every slippery-ice observation.	☐	☐	☐
Frictional heating can contribute during sliding.	☐	☐	☐
One mechanism must be the sole explanation.	☐	☐	☐

The crucial distinction

Complete both sentences.

Pressure melting is real because _____

Pressure melting is insufficient because _____

4. What did reduction buy us?

Tick every genuine gain produced by moving from “ice is slippery” to a quantitative physical model.

- | | |
|--|---|
| ☐ a calculable prediction | ☐ proof that no other mechanism matters |
| ☐ a stated range of conditions | ☐ a way to compare rival accounts |
| ☐ a possible falsifying observation | ☐ the social reason Aunt Minnie is in hospital |

4 Transfer the method: audit a familiar “because”

Four compressed classroom answers

1. **Chalk is white** because calcium carbonate is white.
 2. **The sky is blue** because blue light scatters more.
 3. **Deep water is blue** because water is transparent.
 4. **Leaves are green** because chlorophyll reflects green light.
- Each sentence contains something useful. Each also hides a seam.

1. Choose one case

Chosen case: _____

CLAIM	Rewrite the smallest defensible statement.
Your sentence	_____
WARRANT	What measurement, experiment, or model would support it?
Evidence needed	_____
BOUNDARY	Where, when, or at what scale might it stop working?
Conditions	_____
OPENING	What deeper question does the answer make possible?
Next question	_____

2. Sort the hidden explanatory jobs

Circle every question your present evidence can answer:

observation mechanism history current role purpose

Choose one it cannot yet answer and explain what evidence is missing:

3. Explanation for whom?

How would your stopping point change for each audience?

A seven-year-old asks

A GCSE examiner asks

A researcher asks

Audience changes the stopping point, not the evidence

An explanation can be shorter for a novice, although its claims do not become truer because the audience knows less. Keep a route back to the omitted mechanism, assumptions, and contrary evidence.

4. The reductionism judgement

Complete this sentence with a qualification:

The deeper account is a better explanation when...

5 Final judgement: earn your “because”

Choose one proposition to evaluate

P: “Feynman’s pressure-melting explanation is wrong.”

Q: “The best scientific explanation is always the deepest available reduction.”

Write 160–200 words. Refer to the video and at least two dossier sources by letter.

Your response must:

- state a precise judgement rather than choosing a slogan;
- distinguish the philosophical point of the video from the ice mechanism;
- use one numerical result or specified observation;
- disclose evidence that runs against your preferred account;
- state a boundary and finish with a researchable opening question.

Discernment check

Level	Evidence visible in my response	Check
Reproduction	I accurately state at least one explanation.	☐
Analysis	I separate claim, evidence and mechanism.	☐
Perception	I notice a hedge, boundary, scale or missing link.	☐
Discernment	I decide what deserves weight and justify that decision against adverse evidence.	☐

Exit sentence

A strong explanation does not end curiosity. It closes one question securely enough to expose the next.

The next question I can now ask is: _____

Source trail: Feynman, *BBC Fun to Imagine* (1983); Rosenberg, *Physics Today* 58(12), 50–55 (2005), doi:10.1063/1.2169444; Dash et al., *Reviews of Modern Physics* 78, 695–741 (2006); Weber et al., *J. Phys. Chem. Lett.* 9, 2838–2842 (2018); Canale et al., *Physical Review X* 9, 041025 (2019). Adapted from “The Cheapness of Because”.